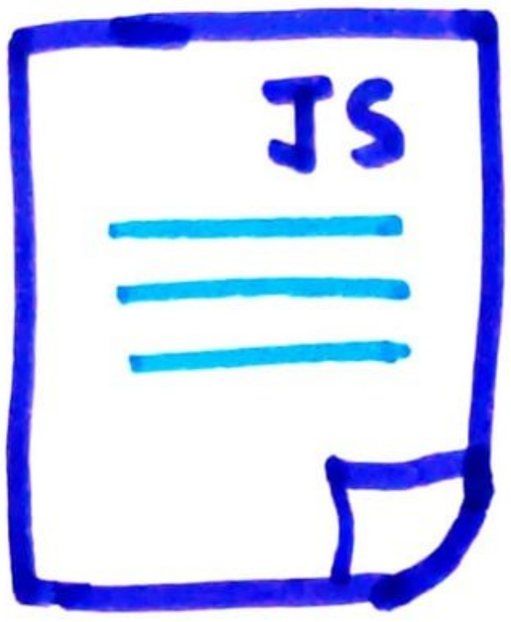
Sold to
pratik501501@gmail.com

JavaScript Engine



Hey, I'm JavaScript Can you help me run



Did someone say anything? I don't understand!

Okay... So the browser doesn't understand JavaScript.

What it understands is bits (1's and 0's)

Who can help us here?

Yes!! The Javascript Engine

codeWithSimran codeWithSimran_

There are a lot of JavaScript Engines out there written by really smart people!

For example :- V8 engine is written in C++ (yes they're programmed too and can be in a different language)

Okay, so what's inside this JavaScript Engine?

JE

Memory heap

This is where all the memory gets allocated e.g. var a=5; memory allocated to variable a

Call Stack

This is where your program executes. It keeps track of where we are in the code

codeWithSimran codeWithSimran_

So ever heard of a memory leak?

A memory heap has limited space. When you have too much of unused memory that you dont free up the spac gets filled.

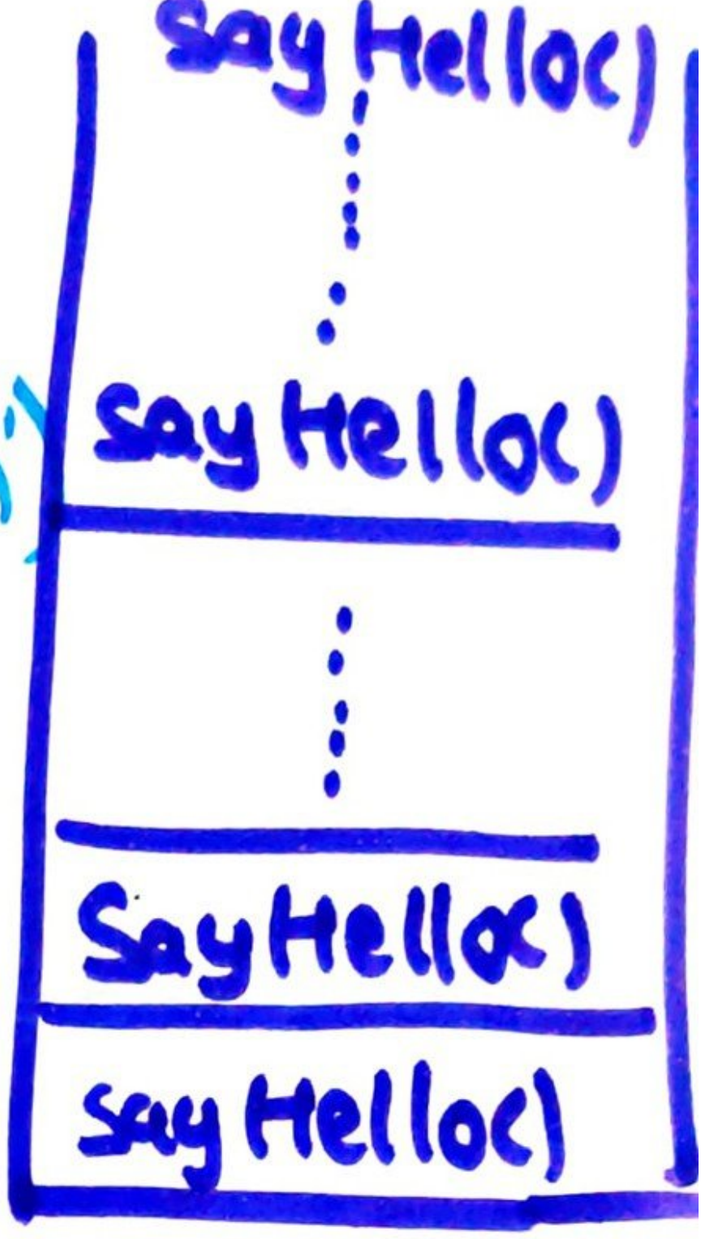
No wonder, global variables are bad (They remain throughout the execution of the code)

You must've heard of stack overflow!!

Well that's when your call Stack overflows as it also has limited spac.

codeWithSimran codeWithSimran_

```
function SayHello()
{
    console.log('Simran')
    SayHello();
}
```



well that went into an infinite recursion and we have stack overflow

JavaScript is a single threaded language?

Well that means it has only ONE CALL STACK and therefore it can only execute one task at a time

Okay But Why single threaded!

It's quite easy and no complications

codeWithSimran codeWithSimran_

Okay... Wait! I've heard of asynchronous programming. If JavaScript can do that, how is it single threaded?

Let's take an example!

```
setTimeout ( () => {
        console.log("setTimeout is asyn"
    }, 2000)
```

→ wait for 2 seconds

setTimeOut is given to us by Web APIs (It gives us various APIs) It's technically not a part of Javascript.



codeWithSimran



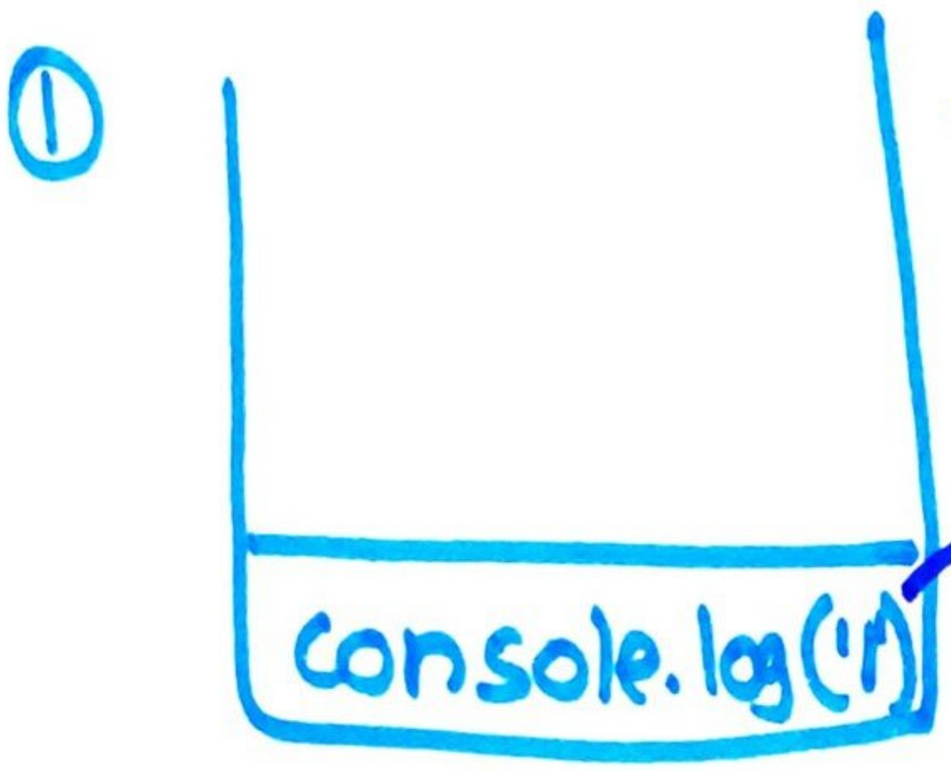
codeWithSimran_

```
console.log('1')
setTimeout(() => {
    console.log('2');
},
console.log('3');
```

Output : 1
3
2

[since setTimeOut waits for 2 secs it's printed in the end]

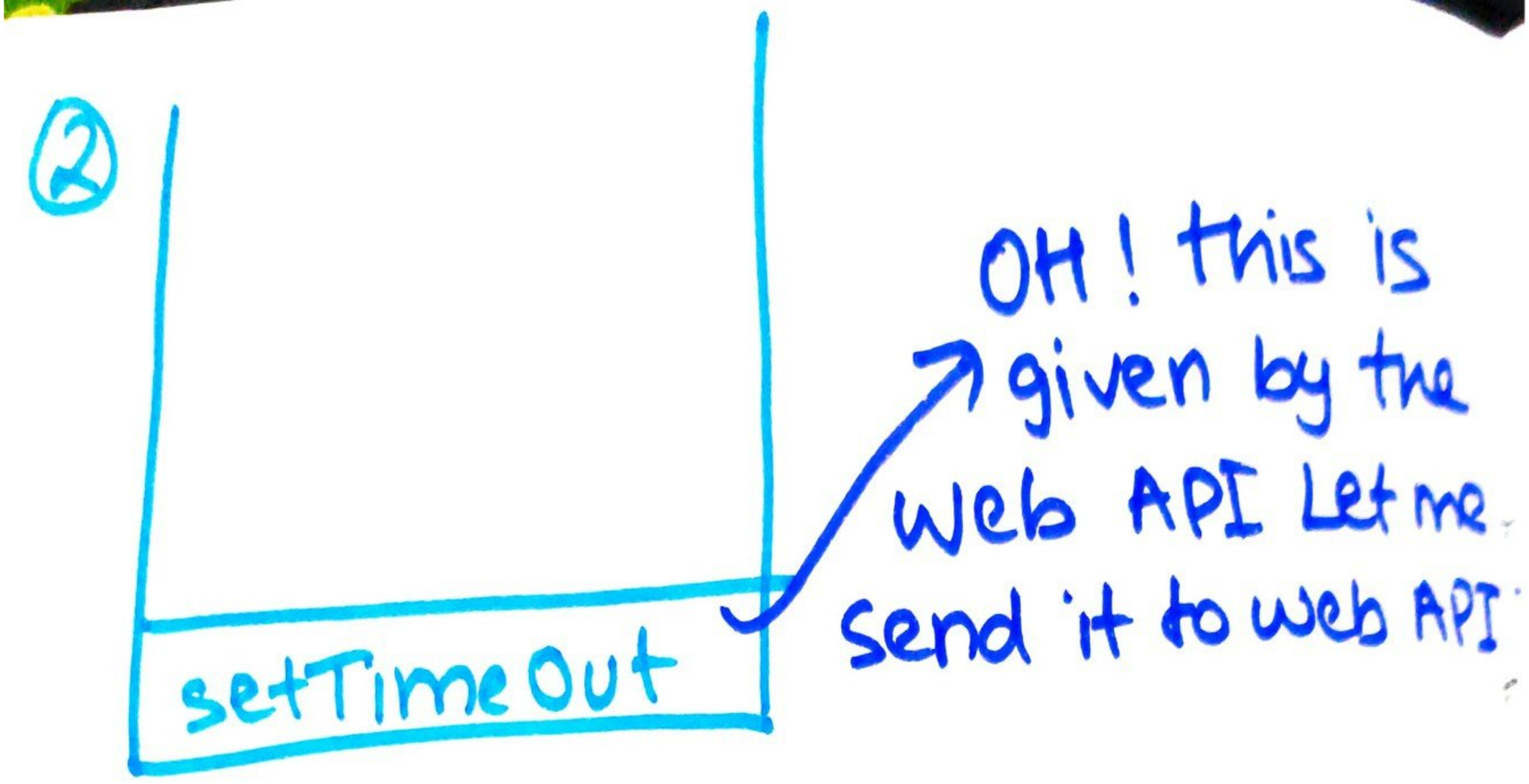
Behind the scenes



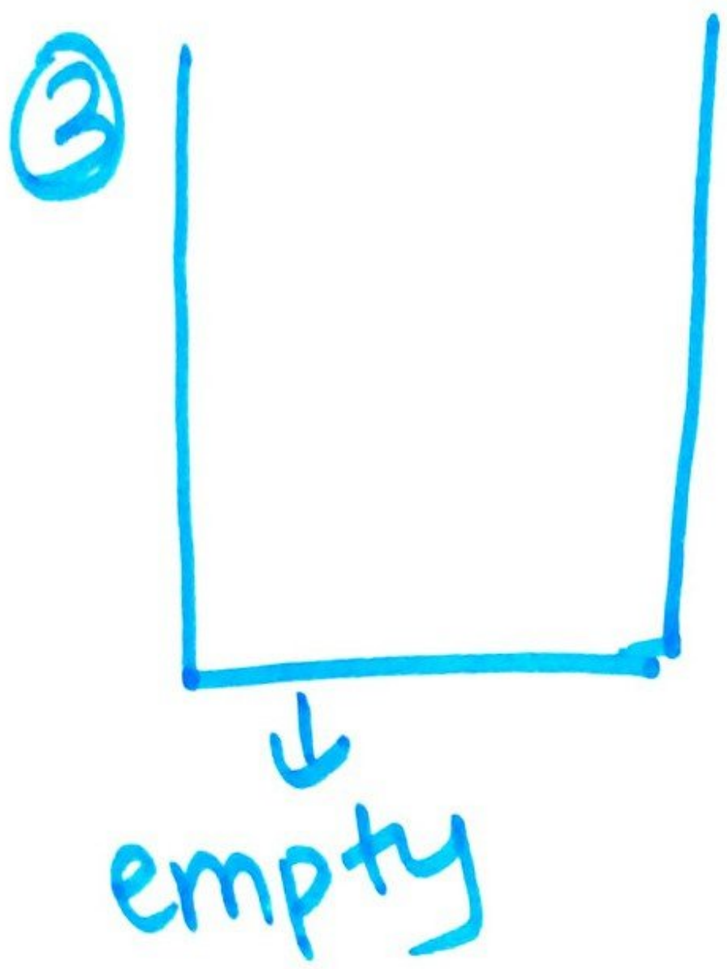
→ Call Stack

→ gets executed and gets popped so the stack is now empty and output is 1

codeWithSimran codeWithSimran_

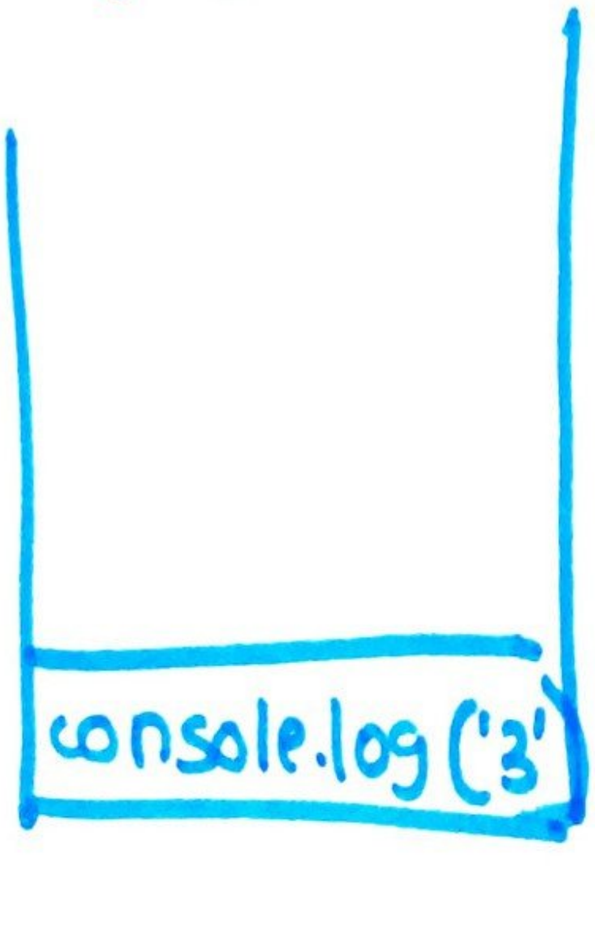


OH! this is given by the web API Let me send it to web API



Web API

I've setTimeout with me and I should execute it after 2 seconds



> Output will print 3

So far

1

3

Web API is still waiting

codeWithSimran codeWithSimran_

(5) After 2 seconds are over WEB API ← On its console.log(2 that should be * executed. This is basically a callback that is executed after 2 secs.

WEB API will send this to callback Queue saying there's a callback please proceed.



callback queue

This queue basically keeps track of all callbacks that need to be executed.

codeWithSimran codeWithSimran_

Now, there's something called as event loops which keeps checking if stack is empty

Well now it's empty so the event loop will take a callback from callback quewe and put it in the stack



> prints 2

so finally we have

1
3
2

Recap of setTimeOut

① Pushed to stack → ② Passed to WEB API

④ Pushes callback to callback queue ← ③ waits for 2 seconds

→ ⑤ Event Loop ckeck if stack empty and pushes to stack

codeWithSimran codeWithSimran_